

- with database 197–201
- with file uploads 206–207
- Text Model class 317
- TextChoices class 163
- TextField 162, 181
- third-party libraries 208–209
 - installing 326–329
- through argument 363
- through_fields argument 363
- title attribute 116
- to_field argument 362
- tool libraries 304–306
 - Django Awl 306
 - Django Extensions 305
- Truncator 165–166, 189
- try block 165
- type attribute 153

U

- UI tests 193
- unique_together 123–124
- uniqueness 123–124
- unit tests 192
- unittest runner 196
- unittest class 197
- unittest framework 194
- unittest module 194, 202, 211
- UPDATE statement 219
- upload_to argument 182
- uploads 152–188
 - configuring project for 178–180
 - file upload views 183–187
 - restricted file downloads 188
 - storing and referencing uploaded files 180–187
 - web forms 153–162
- | upper filter 42, 51
- URL arguments 81–85
- <url> tag 148–149
- url tag 86, 176
- URLField 359
- urlpatterns 23
- urlpatterns variable 9
- URLs, mapping 8–10
- User account object 132
- user data
 - uploads 178–188
 - web forms 153–162
- user management
 - authorization in views 139–142
 - handling logouts 141–142
 - multiuser websites 130

- restricting access to views 137–139
- signals to automatically create user
 - profile 142–145
- storing user data 131–136
 - adding user profile 134–136
 - choosing user account mechanism 132
- User Model object 134, 202
- User object 131, 148, 150
- User.objects.create_superuser() method 210
- UserAdmin class 135
- UserProfile object 135, 144, 183, 186, 199
- users
 - exercise solution 377–379
 - password management 145–150
 - new password form 149–150
 - password reset email 147–149
 - password reset page 147
 - restricting access to views 137–139

V

- validate_templates command 305
- ValidationError 163, 165
- validators 316
- .value() method 117
- Venue 91–93, 104, 124, 134, 150, 181, 183, 185, 202, 206–207
- venue_id 184
- VenueForm 183
- venues management command 229
- venues_controlled 183
- venues_restricted 150
- venues() 185–187
- venv. *See also* virtual environments 325
- verbatim blocks 49
- verbose_name attribute 124
- verbose_name_plural attribute 124
- version argument 282
- version management command 228
- versioning APIs 258
- Video Model class 317
- views 4–5, 23, 33–35
 - authorization in 139–142
 - handling logouts 141–142
 - databases 11–13
 - file upload views 183–187
 - handling GET and POST in 155–160
 - mapping URLs 8–10
 - multiuser sites 15
 - querying models in 81–89
 - listing pages 86
 - using query parameters for pagination 87–89
 - using URL arguments 81–85